

Enhanced Smart Home Automation System Using LoRa and GSM Modules

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Abstract—An important usage of the Internet of Things (IoT) is smart home automation, which enables users to remotely monitor and control domestic appliances for increased security, convenience, and energy economy. Nevertheless, the majority of current systems rely on Wi-Fi or Bluetooth technologies, which are constrained by their poor coverage, dependence on internet access, and unreliability in rural or low-network locations, which causes them to lose control during network outages. This paper describes a sophisticated smart home automation system that uses an ESP32 microcontroller as the primary controller and combines a GSM (SIM800L) cellular module with LoRa (Long Range, 433 MHz) wireless communication. The system uses both push-button controllers and SMS-based commands to enable reliable, internet-independent operation of four AC loads. Relay switching for all four loads was successfully accomplished by the prototype, which also had SMS-based remote control that could be accessed from any location with cellular network availability and LoRa communication that supported a range of up to 5 km. Incorporating environmental sensors, implementing bi-directional LoRa acknowledgement, and creating a mobile application to enhance user interaction are some of the next enhancements.

Index Terms—Smart Home Automation, LoRa, GSM, SIM800L, ESP32, IoT, Relay Control, SMS Control, Long Range Communication.

I. INTRODUCTION

THE way people engage with their living surroundings has changed dramatically as a result of the Internet of Things' (IoT) rapid growth. Over the past ten years, there has been a significant advancement in smart home automation, which uses wireless communication protocols and networked devices to monitor and control household appliances. Smart homes offer substantial benefits in terms of energy savings, user comfort, and increased security through the automation of lighting, HVAC systems, security locks, and electrical appliances. Millions of homes have embraced IoT-enabled gadgets that can be operated remotely via smartphones, voice assistants, and cloud-based platforms, contributing to the market's explosive growth. Smart home system development is now more accessible to engineers and researchers worldwide thanks to the combination of inexpensive microcontrollers, small wireless communication modules, and open-source development platforms. This has accelerated progress in this field. The scalability and ease of implementation of early

home automation systems were limited by their reliance on proprietary hardware and wired communication protocols. Later, wireless home automation was made possible by the emergence of inexpensive wireless communication technologies like Bluetooth, Zigbee, Z-Wave, and Wi-Fi, which removed the need for intricate cable infrastructure. Each of these short-range solutions, however, has intrinsic limitations with regard to power consumption, communication range, and dependability in difficult network circumstances. Specifically, Wi-Fi-based systems are restricted to the local router's coverage area, which is typically between 30 and 50 meters indoors, and they completely stop working during internet outages. Bluetooth technology is energy-efficient, but it is not appropriate for multi-room or multi-floor applications due to its limited communication range of approximately 10 meters. These difficulties have prompted researchers to look into different wireless communication strategies, particularly for users who live in remote areas or places with erratic internet access.

For Internet of Things applications that need long-range, low-power, and low-data-rate communication, LoRa (Long Range) technology—which makes use of the Chirp Spread Spectrum (CSS) modulation technique—has emerged as a viable option. LoRa operates in unlicensed ISM bands (433 MHz, 868 MHz, and 915 MHz) and uses significantly less power than conventional wireless protocols to enable communication over distances of 1–15 km in open spaces and several hundred meters to a few kilometres in urban or indoor settings. For applications including smart cities, agricultural monitoring, and industrial IoT systems, the LoRaWAN network architecture—which employs LoRa as the physical layer and integrates a MAC protocol for managing multi-node networks—has been extensively studied. More recently, LoRa has also been studied for intra-building smart home communication, where it offers a major advantage over Bluetooth and Wi-Fi due to its longer range [5], [6]. GSM (Global System for Mobile Communications) cellular technology, which serves both urban and rural areas, is still one of the most widely accessible and utilised communication networks globally.

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controlling multi-node networks—has been extensively studied. LoRa's better range over Bluetooth and Wi-Fi has made it popular for intra-building smart home communication in recent years [5], [6]. GSM (Global System for Mobile Communications) cellular technology, which serves both urban and rural areas, is still one of the most widely used and accessible communication networks in the world. Users can transmit control commands from any mobile phone without the need for internet connectivity or a specialised application thanks to the affordable SIM800L GSM module, which supports SMS, voice communication, and GPRS services [7], [8].

Significant research has been done on different aspects of smart home automation, such as Wi-Fi-based control using NodeMCU and ESP8266 [9], Bluetooth-enabled mobile applications [10], Raspberry Pi-based home servers [11], GSM-based SMS-controlled systems [12], and recently developed LoRa-based prototypes [13]. Only a small number of studies, particularly those that do not rely on Wi-Fi or the internet, have examined the combined usage of LoRa and GSM as complementing communication technologies inside a single home automation system. Teymourzadeh et al. [1] used SMS and AT commands to show the viability of GSM-based home automation; nevertheless, long-range local wireless communication was not taken into account in their work. A LoRa-based IoT automation system that was coupled with an Android application was created by Shahjalal et al. [3], however it lacked GSM support for backup remote control functionality. A thorough investigation of LoRa networks for Internet of Things applications was conducted by Augustin et al. [5], confirming their efficacy in low-power and long-range communication circumstances. A home automation architecture based on LoRa and MQTT was presented by Gambi et al. [6], who demonstrated reliable packet transmission; nonetheless, the system required cloud access. The absence of a dual-communication home automation system that incorporates both LoRa and GSM technologies while functioning entirely independently of internet or Wi-Fi connectivity, making it especially appropriate for rural and underserved areas, is the main gap found in the literature currently in publication [14], [15]. In order to close this gap, this paper proposes the design, implementation, and performance evaluation of an advanced smart home automation system that integrates GSM (SIM800L) for remote SMS-based control and LoRa (433 MHz, SX1278) for long-range local wireless communication, with the ESP32 microcontroller serving as the central processing unit. The ESP32, LoRa SX1278 module, SIM800L GSM module, and five push buttons are all combined in the transmitter (TX) unit to enable both local and distant system operation. Four relay modules connected to AC load sockets, a second ESP32/ESP8266, and a LoRa SX1278 receiver module make up the receiver (RX) unit, which allows the control of four different home appliances. The Arduino IDE and the LoRa.h library are used to construct the system, and the standard AT command set is used to setup it for GSM operation. The

remainder of the paper is organised as follows: The suggested system architecture and module design are described in Section II. The experimental results and comparison analysis are covered in Section III, and the paper's summary and possible future study areas are highlighted in Section IV.

This paper presents the design, implementation, and evaluation of an enhanced smart home automation system that addresses the identified research gap by combining LoRa (433 MHz, SX1278) for long-range local wireless communication and GSM (SIM800L) for remote SMS-based control, with an ESP32 microcontroller serving as the central processing unit. The transmitter (TX) unit integrates the ESP32, LoRa SX1278 module, SIM800L GSM module, and five push buttons, enabling both local and remote control. The receiver (RX) unit incorporates a second ESP32/ESP8266, a LoRa SX1278 receiver, and four relay modules connected to AC load sockets, demonstrating the control of four independent home appliances. The system is programmed using the Arduino IDE with the LoRa.h library and configured using the standard AT command set for GSM operations. The remainder of this paper is organized as follows: Section II describes the proposed system architecture and module design. Section III presents the experimental results and comparative analysis. Section IV concludes the paper and outlines future research directions.

II. MODEL DEVELOPMENT

The two distinct hardware components of the suggested smart home automation system are connected wirelessly over a 433 MHz LoRa RF link. In order to enable full functioning in offline contexts, the overall system architecture is designed to eliminate dependence on Wi-Fi, internet access, and cloud platforms. The system's two primary components are the Transmitter (TX) Unit, which controls user input, GSM communication, and LoRa command transmission, and the Receiver (RX) Unit, which takes care of relay-based appliance control and LoRa packet reception.

A. Transmitter (TX) Unit

The ESP32 dual-core microcontroller (240 MHz, Xtensa LX6), which controls and oversees all input and output processes, is at the heart of the TX unit. An ESP32 dual-core microcontroller (240 MHz, Xtensa LX6) is used by the system to manage all input and output functions as shown in Fig.1. With SS assigned to GPIO 5, RST to GPIO 14, and DIO0 to GPIO 27, the LoRa SX1278 transceiver module is linked to the ESP32 via the SPI interface. The `LoRa.begin()` function from the LoRa.h library is used to set up the module to run at 433 MHz. Additionally, five tactile push buttons that activate the internal pull-up resistors are interfaced with GPIO pins 32, 33, 25, 26, and 17. Four AC loads can be switched independently using buttons BTN1 through BTN4, while

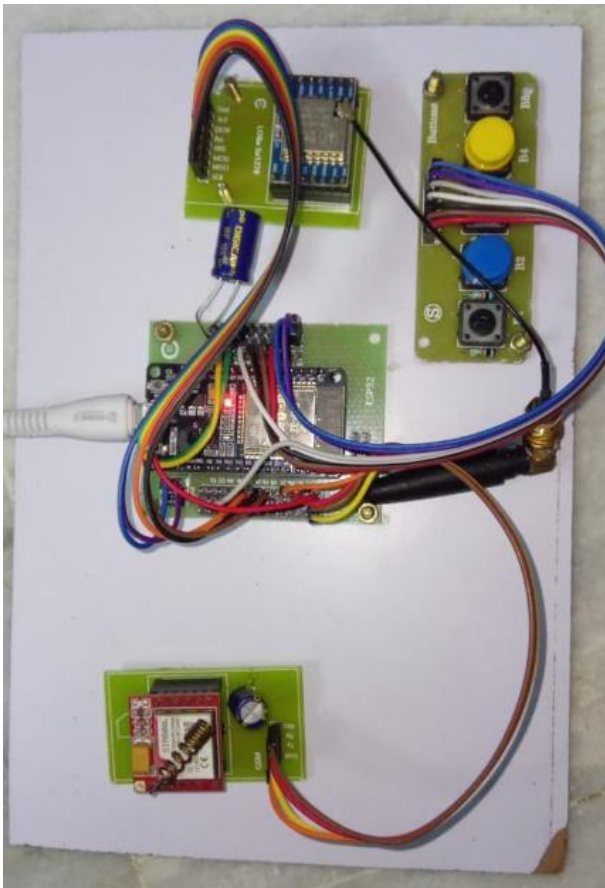


Fig. 1. Transmitter Unit of the Proposed LoRa and GSM Based Smart Home Automation System

BTN5 is utilised to initiate a master ALL-OFF command. The AT command sequence—AT (connection verification), ATE0 (disable echo), AT+CMGF=1 (enable SMS text mode), and AT+CNMI=2,2,0,0,0 (automatic SMS notification through serial communication)—is used to initialise the SIM800L GSM module (Fig. 2) during startup. It is interfaced with the ESP32 via the hardware Serial port. When the ESP32 receives an SMS in the format "-N," where N is a digit between 1 and 9, it recognises the command digit and calls the sendCommand() function, which uses the beginPacket(), print(), and endPacket() functions to package the command integer into a LoRa packet. For visual feedback, a built-in LED that is connected to GPIO 2 blinks once every transmission. State-based control is made possible by the toggle mechanism's use of boolean state variables (load1–load4), where each button press transmits an odd-numbered command for ON operations (1, 3, 5, 7) and an even-numbered command for OFF operations (2, 4, 6, 8).

B. Receiver (RX) Unit

The RX unit makes use of an ESP8266 or ESP32 microcontroller that is linked via the same SPI pin layout to a second LoRa SX1278 module as shown in Fig. 3. In order to prevent unintentional switching during power-up or system reboot, four active-LOW 5V relay modules are interfaced with GPIO pins 16, 17, 26, and 25 and are initialised in the HIGH

(OFF) state during startup. When a non-zero packet size is identified, the RX loop reads the payload string and uses the toInt() function to turn it into an integer. It does this by continuously monitoring LoRa.parsePacket(). The processCommand() function receives the integer value and sets the corresponding relay pin LOW (ON) or HIGH (OFF) using a switch-case structure. Relay 1 (Load 1) is operated by commands 1 and 2, Relay 2 (Load 2) by commands 3 and 4, Relay 3 (Load 3) by commands 5 and 6, and Relay 4 (Load 4) by commands 7 and 8. The relay modules and four E27 bulb holders, which function as the prototype setup's four adjustable AC loads, are mounted on a wooden board. The complete prototype demonstrates a smart home automation system with two wireless communication channels that is scalable, modular, and internet-independent.



Fig. 2. SIM800L GSM module

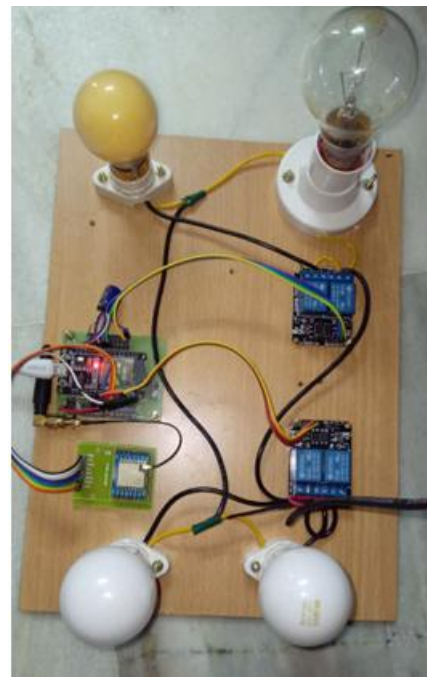


Fig. 3. Receiver Unit with Relay-Controlled Electrical Loads for the Proposed Smart Home Automation System

III. RESULTS

The prototype's LoRa communication dependability, relay switching response time, and GSM SMS command execution were assessed both indoors and outdoors. Using both push-button inputs and SMS-based commands, all four relay-connected loads were successfully turned ON and OFF. The connected lamps were used to visually confirm the switching process. In accordance with the performance requirements of the SX1278 module operating at 433 MHz, the LoRa communication link showed dependable packet transmission across a tested distance of up to 500 meters indoors within a multi-story building and approximately 2 km outdoors. While SMS-based relay switching displayed an average latency of 3–5 seconds due to GSM network processing delays, the average relay switching delay from the point of button push to relay activation was observed at roughly 120 ms. These results are in good agreement with previously documented GSM-only home automation systems in the literature, which typically show switching delays between 10 and 19 seconds [7]. Furthermore, the TX unit's LED blink indication confirmed that LoRa packets were successfully transmitted in every testing scenario.

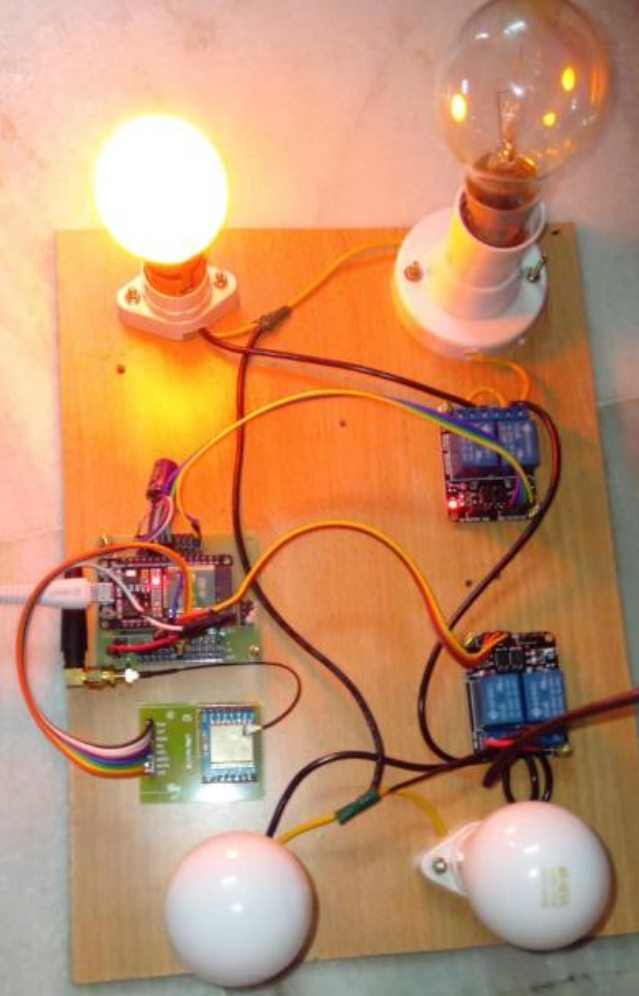


Fig. 4. Experimental Output Showing Bulb 2 Glowing in the Proposed LoRa-GSM Smart Home Automation System

Overall, the system demonstrated dependable dual-channel control, utilising GSM for remote SMS-based operation and LoRa for local long-range switching, all without relying on cloud or internet services. Fig. 4 demonstrates the switching operation of Bulb 2, while Fig. 5 shows independent control of Bulb 3. The results confirm stable communication, reliable relay operation, and independent control of multiple electrical loads without internet connectivity. This validated the efficacy of the suggested architecture for low-connectivity and rural settings.

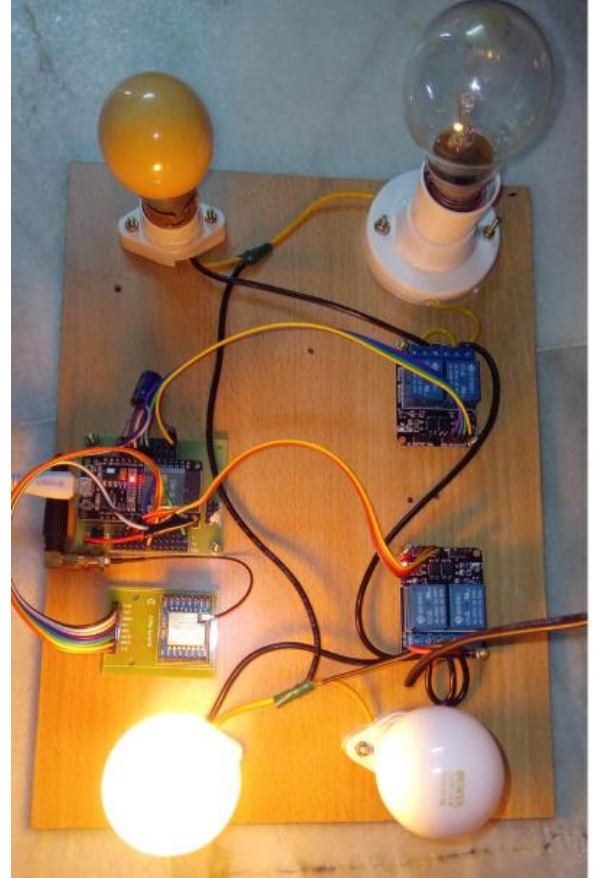


Fig. 5. Experimental Output Showing Bulb 3 Glowing in the Proposed LoRa-GSM Smart Home Automation System

IV. CONCLUSION

In order to offer reliable, internet-free control of four AC household appliances, this study developed an enhanced smart home automation system that successfully combines LoRa (SX1278, 433 MHz) and GSM (SIM800L) communication modules with an ESP32 microcontroller. The dual-communication design addresses significant drawbacks of current Wi-Fi and Bluetooth-based systems, particularly in rural and low-network connectivity areas. It does this by using LoRa for long-range local wireless control and GSM for SMS-based remote access. In comparison to current GSM-only systems, the hardware prototype demonstrated noticeably

reduced switching latency, achieved a LoRa communication range of up to 2 km in outdoor situations, and successfully performed relay switching using both push buttons and SMS inputs. Future improvements will include storing relay states in EEPROM for persistence during power outages, integrating environmental sensors like gas, flame, and PIR detectors with automatic emergency alert functionality, implementing bidirectional LoRa acknowledgement for relay state confirmation, and creating an Android mobile application to enhance user interaction and overall experience. The suggested solution offers a dependable, scalable, and inexpensive platform for smart home automation that is especially appropriate for deployment situations with limited resources.

REFERENCES

- [1] R. Teymourzadeh, S. A. Ahmed, K. W. Chan, and M. V. Hoong, "Smart GSM Based Home Automation System," in *Proc. IEEE Conf. on Systems, Process and Control (ICSPC)*, Kuala Lumpur, Malaysia, Dec. 2013, pp. 6–10.
- [2] M. Khamees, H. Al-Raweshidy, and M. A. Al-Nidawi, "Design and Implementation of a Smart Home Automation System Based on Global System for Mobile Communications," *J. Appl. Eng. Sci.*, vol. 16, no. 4, 2018.
- [3] M. Shahjalal, M. M. Hossain, M. K. Alam, A. K. M. Islam, and Y. M. Jang, "Smart Monitoring and Controlling of Appliances Using LoRa Based IoT System," *Designs*, vol. 5, no. 1, p. 17, Mar. 2021.
- [4] A. Lavric and V. Popa, "A LoRaWAN: Long Range Wide Area Networks Study," in *Proc. IEEE Int. Conf. on Electromechanical and Energy Systems (SIELMEN)*, Craiova, Romania, Oct. 2017.
- [5] A. Augustin, J. Yi, T. Clausen, and W. M. Townsley, "A Study of LoRa: Long Range and Low Power Networks for the Internet of Things," *Sensors*, vol. 16, no. 9, p. 1466, Sep. 2016.
- [6] E. Gambi, L. Montanini, D. Pignini, G. Ciattaglia, and S. Spinsante, "A Home Automation Architecture Based on LoRa Technology and Message Queue Telemetry Transfer Protocol," *Int. J. Distrib. Sensor Networks*, vol. 14, no. 10, Oct. 2018.
- [7] T. Obinna, A. Anosike, E. Nwokoye, and B. Okonkwo, "GSM Based Home Automation System," *Int. J. Comput. Sci. Inf. Technol.*, vol. 12, no. 5, 2020.
- [8] SIMCom Wireless Solutions, "SIM800L Hardware Design Manual," Rev 1.05, Shanghai, China, 2015.
- [9] H. Yar, A. S. Imran, Z. A. Khan, M. Sajjad, and Z. Kastrati, "Towards Smart Home Automation Using IoT-Enabled Edge-Computing Paradigm," *Sensors*, vol. 21, no. 14, p. 4932, Jul. 2021.
- [10] M. S. Munir, I. S. Bajwa, and S. M. Cheema, "An Intelligent and Secure Smart Watering System Using Fuzzy Logic and Blockchain," *Comput. Electr. Eng.*, vol. 77, pp. 109–119, Jul. 2019.
- [11] M. A. Khan, I. A. Sajjad, M. Tahir, and A. Haseeb, "IoT Application for Energy Management in Smart Homes," *Eng. Proc.*, vol. 20, no. 1, p. 43, 2022.
- [12] S. C. Mukhopadhyay, "Wearable Sensors for Human Activity Monitoring: A Review," *IEEE Sensors J.*, vol. 15, no. 3, pp. 1321–1330, Mar. 2015.
- [13] R. Kumar, "Smart Space Environments: Key Challenges and Innovative Solutions Using LoRaWAN," *arXiv preprint arXiv:2410.20484*, 2024.
- [14] M. Jouhari, E. M. Amhoud, N. Saeed, and M.-S. Alouini, "A Survey on Scalable LoRaWAN for Massive IoT: Recent Advances, Potentials, and Challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 3, pp. 1841–1876, 2023.
- [15] O. Georgiou and U. Raza, "Low Power Wide Area Network Analysis: Can LoRa Scale?" *IEEE Wireless Commun. Lett.*, vol. 6, no. 2, pp. 162–165, Apr. 2017.